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Math Maestro

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JEE Mains 2020 Jan Chapter wise Question Bank

Statistics

Q1

If variance of first n natural numbers is 10 and variance of first m even natural numbers is 16 then the value of $m + n$ is

7th Jan Morning

Sol

18

$$\text{Var}(1, 2, \dots, n) = 10 \Rightarrow \frac{1^2 + 2^2 + \dots + n^2}{n} - \left(\frac{1+2+\dots+n}{n} \right)^2 = 10$$

$$\Rightarrow \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2} \right)^2 = 10$$

$$\Rightarrow n^2 - 1 = 120 \Rightarrow n = 11$$

$$\text{Var}(2, 4, 6, \dots, 2m) = 16 \Rightarrow \text{var}(1, 2, \dots, m) = 4$$

$$\Rightarrow m^2 - 1 = 48 \Rightarrow m = 7 \Rightarrow m + n = 18$$

02

If mean and variance of 2, 3, 16, 20, 13, 7, x , y are 10 and 25 respectively then find xy

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Sol

124

$$\text{mean माध्य} = \bar{x} = \frac{2+3+16+20+13+7+x+y}{8} = 10 \Rightarrow x+y = 19 \dots (i)$$

$$\text{variance प्रसरण } \sigma^2 = \frac{\sum (x_i)^2}{8} - (\bar{x})^2 = 25$$

$$\frac{4+9+256+400+169+49+x^2+y^2}{8} - 100 = 25$$

$$\Rightarrow x^2 + y^2 = 113 \dots (ii)$$

$$(x+y)^2 = (19)^2 \Rightarrow x^2 + y^2 + 2xy = 361 \Rightarrow xy = 124$$

(exact data is not retrieved so ans. can vary)

03

Mean and standard deviations of 10 observations are 20 and 2 respectively. If p ($p \neq 0$) is multiplied to each observation and then q ($q \neq 0$) is subtracted then new mean and standard deviation becomes half of original value. Then find q

10 परीक्षणों का समान्तर माध्य एवं मानक विचलन क्रमशः 20 एवं 2 है। यदि प्रत्येक परीक्षण को p ($p \neq 0$) से गुणा करके इनमें से q ($q \neq 0$) घटाया जाए तब नया समान्तर माध्य एवं मानक विचलन वास्तविक मानों के आधे हो जाते हैं, तब q का मान है।

(1) -10

(2) -20

(3) -5

(4) 10

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(2)

If each observation is multiplied with p & then q is subtractedयदि प्रत्येक परीक्षण को p से गुणा करके इनमें से q घटाया जाएNew mean $\bar{x}_1 = p\bar{x} - q$ नया माध्य $\bar{x}_1 = p\bar{x} - q$ $\Rightarrow 10 = p(20) - q$ (1)

and new standard deviations

तथा नया मानक विचलन

$$\sigma_2 = |p| \sigma_1 \Rightarrow 1 = |p| (2) \Rightarrow |p| = \frac{1}{2} \Rightarrow p = \pm \frac{1}{2}$$

$$\text{If } p = \frac{1}{2}$$

$$\text{यदि } p = \frac{1}{2}$$

then $q = 0$ (from equation (1))तब $q = 0$ (समीकरण (1) से)

$$\text{If } p = -\frac{1}{2}$$

$$\text{यदि } p = -\frac{1}{2}$$

$$q = -20$$

04

Mean and variance of 20 observation are 10 and 4. It was found, that in place of 11, 9 was taken by mistake find correct variance.

20 आंकड़ों का माध्य तथा चरिता क्रमशः 10 तथा 4 है। यदि 11 के स्थान पर गलती से 9 लिया गया है तो सही चरिता है—

(1) 3.99

(2) 3.98

(3) 4.01

(4) 4.02

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(1)

$$\frac{\sum x_i}{20} = 10 \quad \dots\dots(i)$$

$$\frac{\sum x_i^2}{20} - 100 = 4 \quad \dots\dots(ii)$$

$$\sum x_i^2 = 104 \times 20 = 2080$$

$$\text{Actual mean सही माध्य} = \frac{200 - 9 + 11}{20} = \frac{202}{20}$$

$$\text{Variance चरिता} = \frac{2080 - 81 + 121}{20} - \left(\frac{202}{20} \right)^2$$

$$= \frac{2120}{20} - (10.1)^2 = 106 - 102.01 = 3.99$$

05

For observations x_i given $\sum_{i=1}^{10} (x_i - 5) = 10$ and $\sum_{i=1}^{10} (x_i - 5)^2 = 40$. If mean and variance of observations

$(x_1 - 3), (x_2 - 3), \dots, (x_{10} - 3)$ is λ & μ respectively then ordered pair (λ, μ) is

आंकड़ों x_i के लिये दिया है कि $\sum_{i=1}^{10} (x_i - 5) = 10$ तथा $\sum_{i=1}^{10} (x_i - 5)^2 = 40$ यदि आंकड़ों

$(x_1 - 3), (x_2 - 3), \dots, (x_{10} - 3)$ का माध्य λ तथा चरिता μ है तो क्रमित युग्म (λ, μ) है-

(1) (3, 3)

(2) (1, 3)

(3) (3, 1)

(4) (1, 1)

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Sol

(1)

$$\text{Mean माध्य } (x_i - 5) = \frac{\sum (x_i - 5)}{10} = 1$$

$$\therefore \lambda = \{\text{Mean माध्य } (x_i - 5)\} + 2 = 3$$

$$\mu = \text{var चरिता } (x_i - 5) = \frac{\sum (x_i - 5)^2}{10} - \frac{\sum (x_i - 5)}{10} = 3$$

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